

SEINAJOKI, Finland

WMO: 029245

Lat: 62.692N

Lon: 22.832E

Elev: 301

StdP: 14.54

Time Zone: 2.00 (EUE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.5	-7.7	-18.6	2.0	-14.0	-11.6	3.0	-7.5	20.5	23.2	18.8	23.1	1.9	130	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	82.0	65.1	77.4	63.1	73.7	61.6	67.6	76.6	65.4	74.0	63.4	71.5	7.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	91.9	71.4	62.4	85.2	69.3	60.4	79.3	66.8	32.2	77.1	30.3	74.1	28.9	71.6	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	15.6	13.8	DB	-19.5	83.7	5.4	5.6	-23.4	87.7	-26.5	91.0	-29.5	94.2	-33.4	98.3	(4)
(5)			WB	-19.6	68.9	4.9	3.4	-23.1	71.4	-26.0	73.4	-28.7	75.3	-32.3	77.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.3	19.1	20.1	25.5	37.6	47.9	55.3	61.2	57.9	49.6	38.9	31.6	25.5	(6)	
	DBStd	16.79	12.79	12.38	10.26	6.02	7.62	6.31	6.07	5.58	6.24	7.35	8.62	10.82	(7)	
	HDD50	4823	957	836	761	373	129	18	0	6	83	348	551	761	(8)	
	HDD65	9438	1422	1256	1226	821	531	298	153	231	463	810	1001	1226	(9)	
	CDD50	915	0	0	0	2	65	178	347	249	70	4	0	0	(10)	
	CDD65	55	0	0	0	0	2	8	35	10	0	0	0	0	(11)	
	CDH74	788	0	0	0	0	46	125	504	113	0	0	0	0	(12)	
	CDH80	158	0	0	0	0	4	26	112	16	0	0	0	0	(13)	
	WSAvg	6.1	6.3	6.3	6.6	6.3	6.4	6.4	5.4	5.2	5.7	5.9	6.3	6.4	(14)	
	PrecAvg	22.0	1.5	1.1	1.1	0.9	1.7	2.4	3.3	2.6	2.0	2.2	1.8	1.5	(15)	
(16) Precipitation	PrecMax	26.3	2.6	2.0	2.1	3.1	3.1	4.0	5.3	4.4	5.2	4.7	4.0	2.5	(16)	
	PrecMin	18.2	0.4	0.0	0.2	0.0	0.4	0.8	0.8	0.7	0.8	0.6	0.2	0.5	(17)	
	PrecStd	2.6	0.6	0.6	0.5	0.7	0.9	0.9	1.4	1.0	1.2	1.1	0.9	0.7	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.8	42.7	51.7	67.6	79.3	84.0	87.7	82.4	69.9	57.5	50.3	44.3	(19)	
		MCWB	39.4	38.5	41.9	51.2	59.6	64.8	67.5	65.5	59.4	52.9	48.9	43.5	(20)	
	2%	DB	37.6	39.0	44.9	59.4	73.6	77.2	83.8	76.9	66.2	53.8	46.8	41.2	(21)	
		MCWB	36.7	36.9	38.9	47.6	57.8	62.1	66.9	63.0	58.3	51.1	45.8	40.1	(22)	
	5%	DB	35.7	35.9	41.2	55.1	68.3	72.0	80.2	73.3	62.9	51.8	44.7	39.3	(23)	
		MCWB	35.0	35.0	36.5	44.4	55.0	59.3	64.5	61.9	56.5	49.5	43.8	38.2	(24)	
	10%	DB	33.9	34.1	37.8	50.2	64.1	68.4	76.6	69.7	60.6	49.8	42.6	37.4	(25)	
		MCWB	33.4	33.3	34.7	42.4	52.4	57.8	63.8	60.7	55.6	48.3	41.9	36.7	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	39.0	43.5	53.0	62.1	67.5	71.1	69.8	61.6	54.3	49.1	43.1	(27)	
		MCDB	39.9	41.0	49.9	63.0	73.6	76.3	82.3	77.7	67.0	57.1	50.3	43.3	(28)	
	2%	WB	36.7	36.6	39.5	48.7	59.4	64.1	68.8	65.7	59.5	52.1	46.1	40.2	(29)	
		MCDB	37.1	38.0	43.5	57.9	71.5	75.0	79.2	72.3	64.3	53.8	46.8	40.8	(30)	
	5%	WB	34.7	35.0	37.2	45.6	56.6	61.4	66.8	63.5	57.7	50.1	43.5	38.0	(31)	
		MCDB	35.2	35.9	40.2	52.6	66.4	70.2	76.0	69.8	61.8	51.4	44.2	38.6	(32)	
	10%	WB	33.1	33.4	35.1	42.6	53.2	59.1	64.8	61.5	56.0	47.9	41.3	36.1	(33)	
		MCDB	33.5	33.9	37.3	49.4	62.1	66.5	73.7	67.4	59.7	49.0	41.8	36.7	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	11.1	12.3	16.3	17.0	20.9	19.6	20.6	19.2	16.6	11.0	7.7	9.2	(35)	
		MCDBR	8.3	9.9	16.7	27.3	29.6	26.9	28.2	25.7	20.9	13.0	8.4	7.8	(36)	
		MCWBR	8.0	8.7	11.7	16.4	15.3	13.8	13.0	13.8	13.7	10.2	7.9	7.6	(37)	
	5% WB	MCDBR	7.9	9.3	15.1	24.5	25.3	23.5	23.0	20.9	18.0	11.0	8.2	7.8	(38)	
		MCWBR	7.8	8.4	11.1	15.7	13.9	13.0	11.7	12.2	12.4	9.5	8.0	7.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.236	0.291	0.316	0.356	0.352	0.350	0.373	0.358	0.334	0.316	0.257	0.352	(40)	
	taud		2.061	2.181	2.268	2.343	2.429	2.455	2.412	2.445	2.507	2.393	2.147	2.597	(41)	
	Ebn at Noon		130	205	249	262	275	277	266	260	246	200	121	77	(42)	
	Edh at Noon		10	20	27	31	31	31	32	28	22	17	10	6	(43)	
(44) All-Sky Solar Radiation	RadAvg		58	238	716	1172	1579	1707	1601	1225	723	287	73	28	(44)	
	RadStd		8	52	74	138	168	149	156	129	82	59	13	3	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page